

Installation of Splunk SIEM and Log Forwarding

Introduction to the Lab

Splunk is an innovative technology which searches and indexes log files and helps organizations derive insights from the data. A main benefit of Splunk is that it uses indexes to store data, and so does not require a separate database to store its information. Splunk is used for monitoring and searching through big data. It indexes and correlates information in a container that makes it searchable, and makes it possible to generate alerts, reports, and visualizations. It can recognize data patterns, create metrics, and help diagnose problems, for business challenges like IT management, security, and compliance.

Splunk forwarders consume data and send it to an indexer. Forwarders require minimal resources and have little impact on performance, so they can usually reside on the machines where the data originates. There are two kinds of forwarders in Splunk.

- i) The **universal forwarder** contains only the components that are necessary to forward data. The universal forwarder is designed to run on production servers, having minimal CPU and memory usage and the least impact possible on mission-critical software.
- ii) A **heavy forwarder is a full Splunk Enterprise instance** that can index, search, and change data as well as forward it. Splunk forwarder in our Splunk server ubuntu machine acts as an agent for log collection from remote machines. Splunk forwarder collects logs from remote machines

and forwards them to the indexer (Splunk database) for further processing and storage.

Note: Universal forwarders do not have a web or application interface. Once installed, you must make configuration changes at the command line in both Windows and Unix- or Linux-based systems.

Best practices:

- Use the universal forwarder when possible as a data collection method.
- Stop and start the universal forwarder from the command line.

In this lab, you will install Splunk SIEM as a standalone machine on a Linux Ubuntu OS. You will be configuring Universal Forwarder on another Ubuntu machine for sending authentication logs to Splunk server. These logs would be captured in Splunk server.

Pre-requisites of the lab

The pre-requisites (recommended system requirement) for configuring Universal Forwarder and Splunk Server for log collection are:

- VMware Fusion or Workstation (It must be registered; trials will not work)
- Install two Ubuntu Virtual machines
- RAM 2+GB
 - 2 GHz dual core processor or higher
 - 4 GB system memory
 - 20 GB of free hard drive space
 - Internet access
- Make sure both virtual machines are on the same network, Preferably NAT

Steps of the lab:

To get our VM up and running, we will need to execute the following steps:

Step 1: Preparing the first Ubuntu VM for Splunk Installation

Step 2: Configuring receiving port at Ubuntu Splunk Server

Step 3: Download and install Splunk Universal Forwarder on second ubuntu machine

Step 4: Searching and Reporting App – Capture the logs in Splunk Ubuntu Server

Execution of the lab (Saving your screenshots here)

Step 1: Preparing the first Ubuntu VM for Splunk Installation

Open the command terminal and run the following commands

a) Change to the Downloads directory using the command `cd Downloads/`

b) Run the command below to download Splunk.

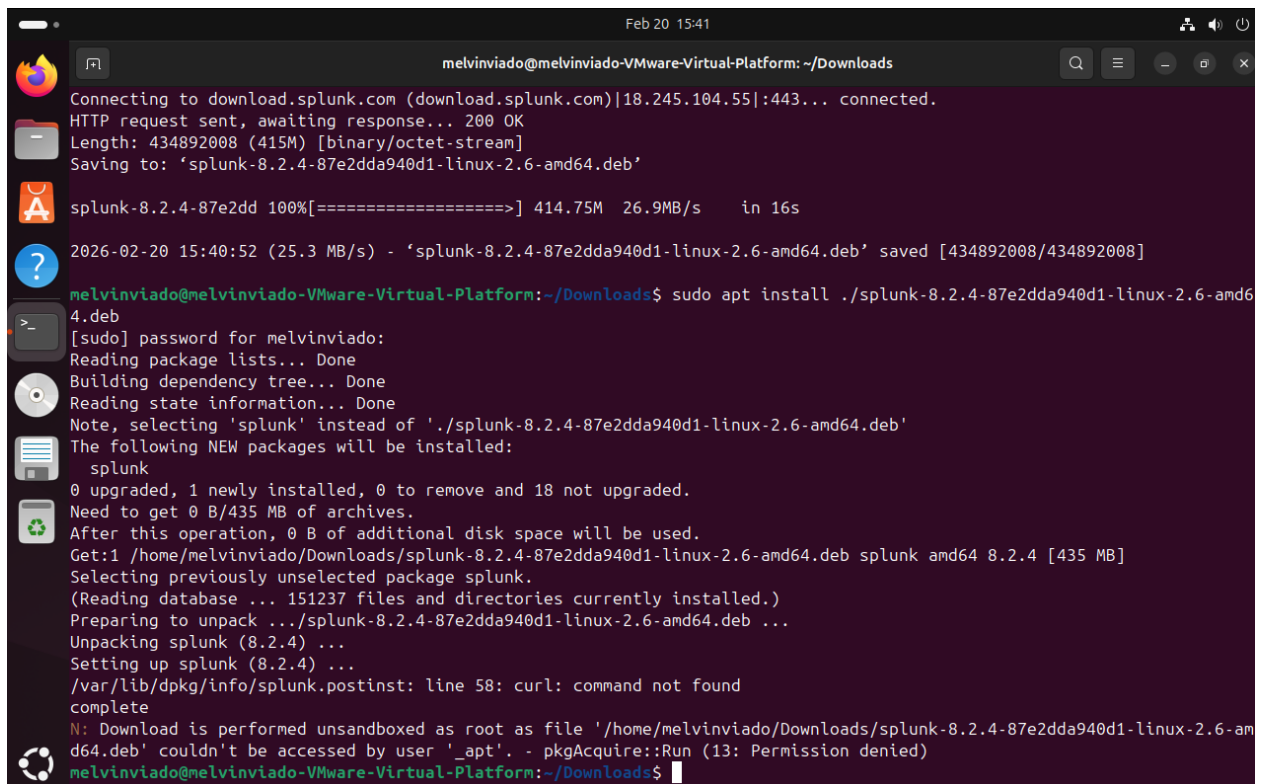
```
wget -O splunk-8.2.4-87e2dda940d1-linux-2.6-amd64.deb
```

```
'https://download.splunk.com/products/splunk/releases/8.2.4/linux/splunk-8.2.4-87e2dda940d1-linux-2.6-amd64.deb'
```

c) Run the command below to install the downloaded Splunk package

```
sudo apt install ./splunk-8.2.4-87e2dda940d1-linux-2.6-amd64.deb
```

[Attach your Screenshot]



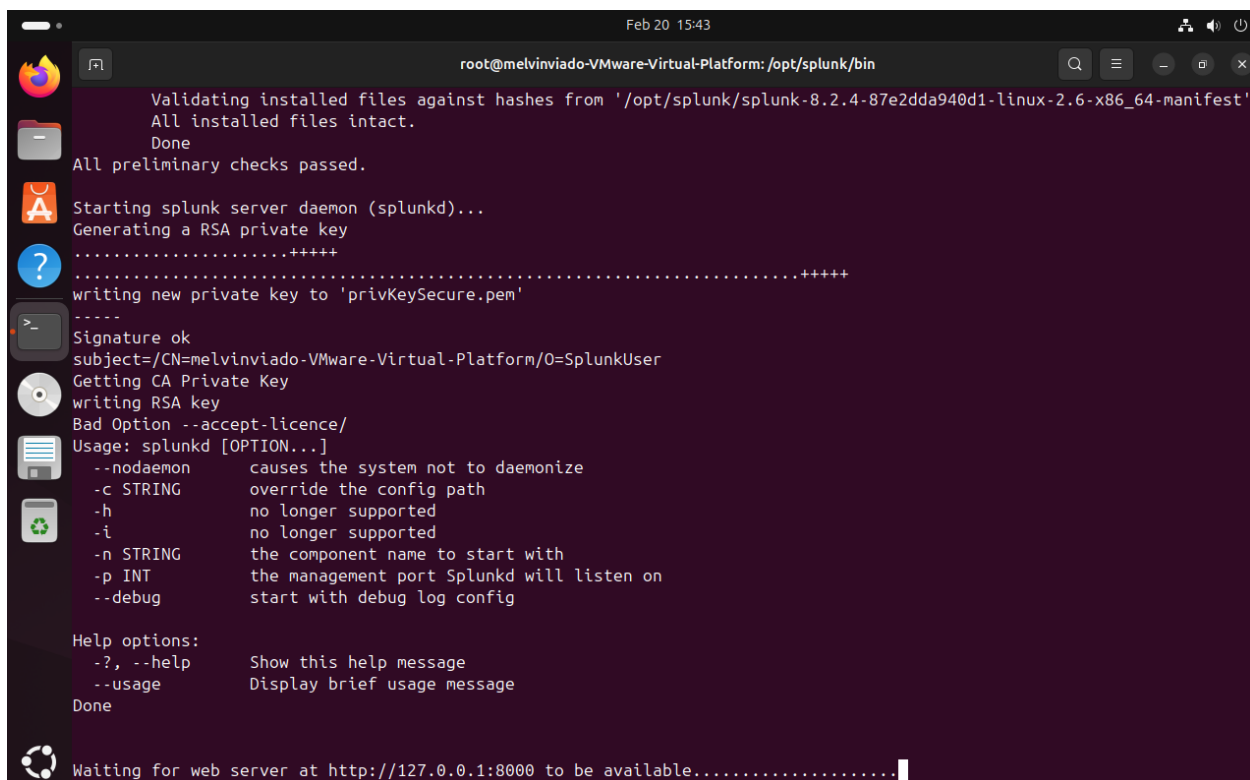
```
melvinviado@melvinviado-VMware-Virtual-Platform: ~/Downloads
Connecting to download.splunk.com (download.splunk.com)|18.245.104.55|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 434892008 (415M) [binary/octet-stream]
Saving to: 'splunk-8.2.4-87e2dda940d1-linux-2.6-amd64.deb'

splunk-8.2.4-87e2dd 100%[=====] 414.75M  26.9MB/s   in 16s

2026-02-20 15:40:52 (25.3 MB/s) - 'splunk-8.2.4-87e2dda940d1-linux-2.6-amd64.deb' saved [434892008/434892008]

melvinviado@melvinviado-VMware-Virtual-Platform: ~/Downloads$ sudo apt install ./splunk-8.2.4-87e2dda940d1-linux-2.6-amd64.deb
[sudo] password for melvinviado:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'splunk' instead of './splunk-8.2.4-87e2dda940d1-linux-2.6-amd64.deb'
The following NEW packages will be installed:
  splunk
0 upgraded, 1 newly installed, 0 to remove and 18 not upgraded.
Need to get 0 B/435 MB of archives.
After this operation, 0 B of additional disk space will be used.
Get:1 /home/melvinviado/Downloads/splunk-8.2.4-87e2dda940d1-linux-2.6-amd64.deb splunk amd64 8.2.4 [435 MB]
Selecting previously unselected package splunk.
(Reading database ... 151237 files and directories currently installed.)
Preparing to unpack .../splunk-8.2.4-87e2dda940d1-linux-2.6-amd64.deb ...
Unpacking splunk (8.2.4) ...
Setting up splunk (8.2.4) ...
/var/lib/dpkg/info/splunk.postinst: line 58: curl: command not found
complete
N: Download is performed unsandboxed as root as file '/home/melvinviado/Downloads/splunk-8.2.4-87e2dda940d1-linux-2.6-amd64.deb' couldn't be accessed by user '_apt'. - pkgAcquire::Run (13: Permission denied)
melvinviado@melvinviado-VMware-Virtual-Platform: ~/Downloads$
```

- d) Switch to root using the command `sudo su` and input your password
- e) Switch to the Splunk directory using the command `cd /opt/splunk/bin`
- f) Run the command to accept Splunk license and
`./splunk start --accept-licence/`
- g) Press enter to get to the end of the License agreement and type y (yes) to accept the license
- h) Input an administrator username with the format `Firstname-NETW2200` and a password you can remember as you won't be able to access Splunk without this. **[Attach your Screenshot]**



```
Feb 20 15:43
root@melvinviado-VMware-Virtual-Platform: /opt/splunk/bin

Validating installed files against hashes from '/opt/splunk/splunk-8.2.4-87e2dda940d1-linux-2.6-x86_64-manifest'
All installed files intact.
Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
Generating a RSA private key
.....+++++
writing new private key to 'privKeySecure.pem'
-----
Signature ok
subject=/CN=melvinviado-VMware-Virtual-Platform/O=SplunkUser
Getting CA Private Key
writing RSA key
Bad Option --accept-licence/
Usage: splunkd [OPTION...]
  --nodaemon      causes the system not to daemonize
  -c STRING       override the config path
  -h              no longer supported
  -i              no longer supported
  -n STRING       the component name to start with
  -p INT          the management port Splunkd will listen on
  --debug         start with debug log config

Help options:
  -?, --help      Show this help message
  --usage         Display brief usage message
Done

Waiting for web server at http://127.0.0.1:8000 to be available.....
```

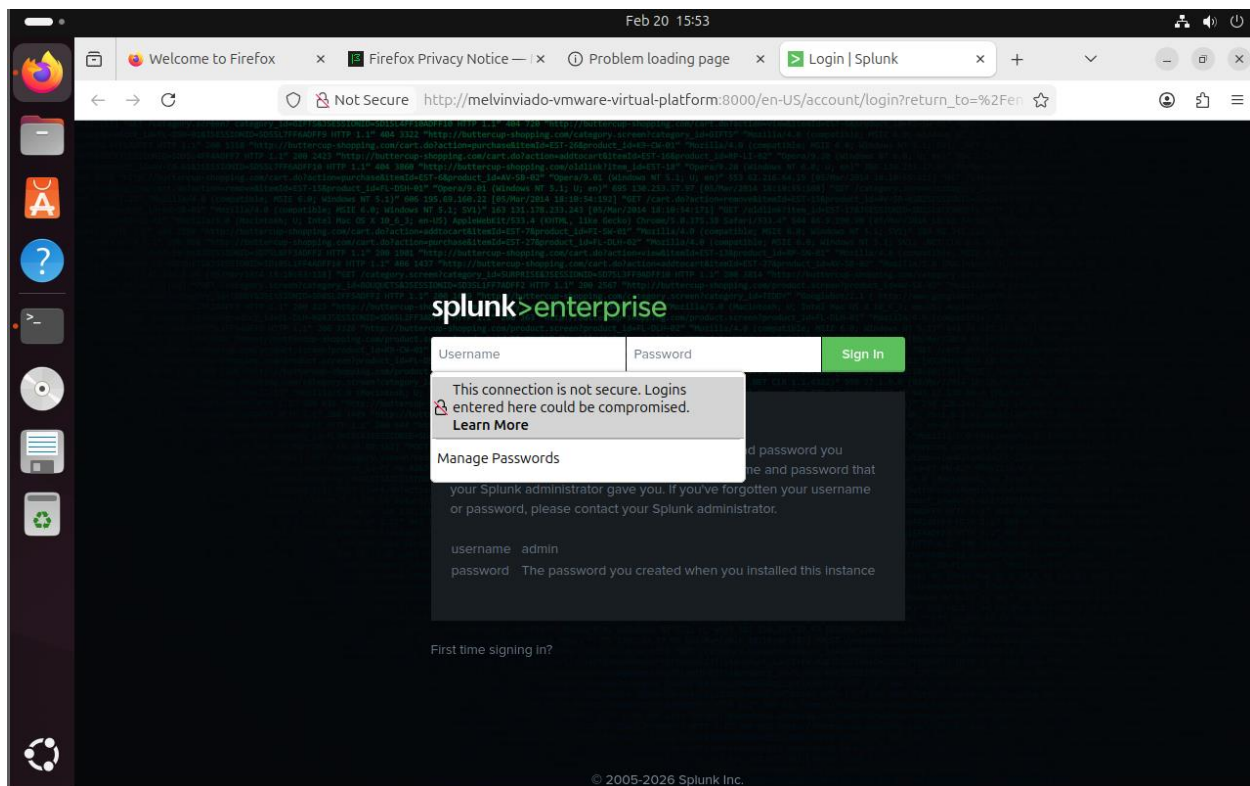
And wait for some time till it's done installing (Sometimes may work right away and some time may get errors)

i) Splunk should be ready and the web url would be in your command line.

Copy the web url and paste on your browser to access Splunk.

Alternatively, Splunk can also be accessed in the browser using the ubuntu Virtual machines IP address routed through port 8000 i.e., 192.168.80.130:8000.

[Attach your Screenshot]

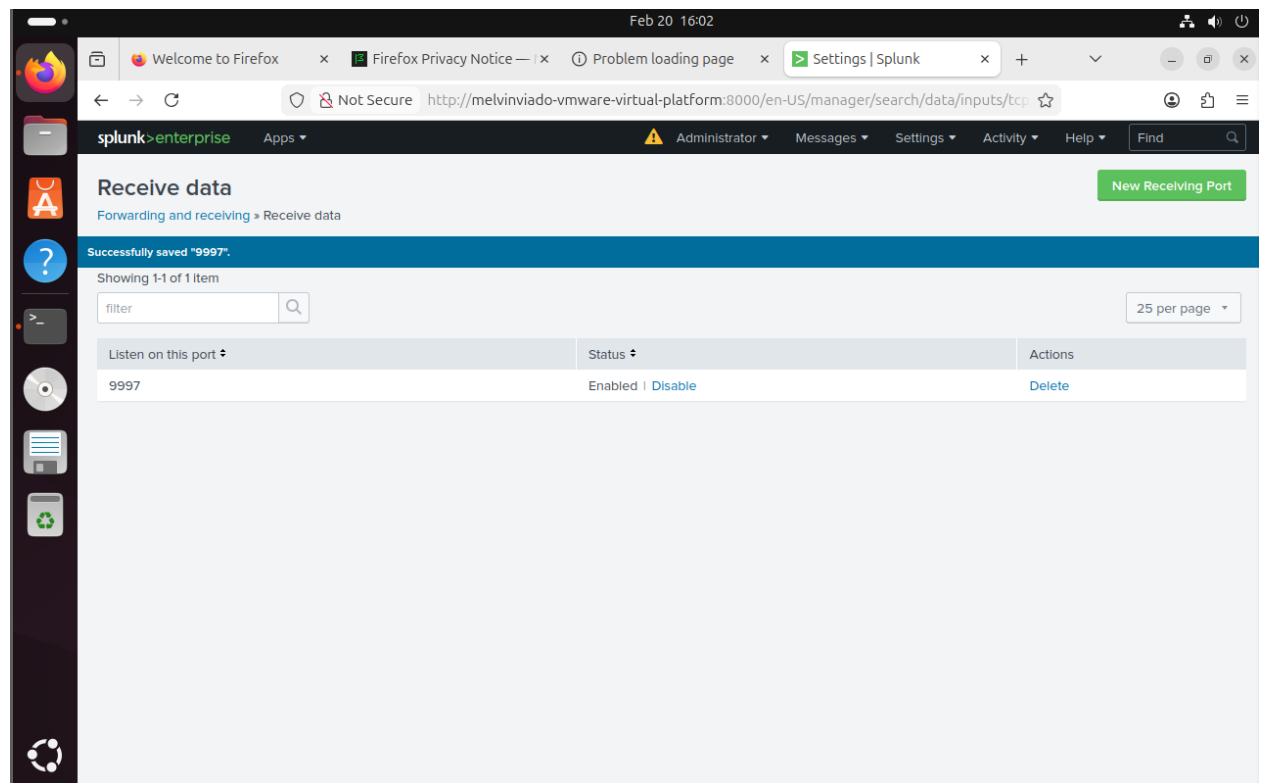


j) Go back to the command line and type the command below to ensure Splunk services starts at any reboot of our virtual machine `./splunk enable boot-start`

Step 2: Configuring receiving port at Ubuntu Splunk Server

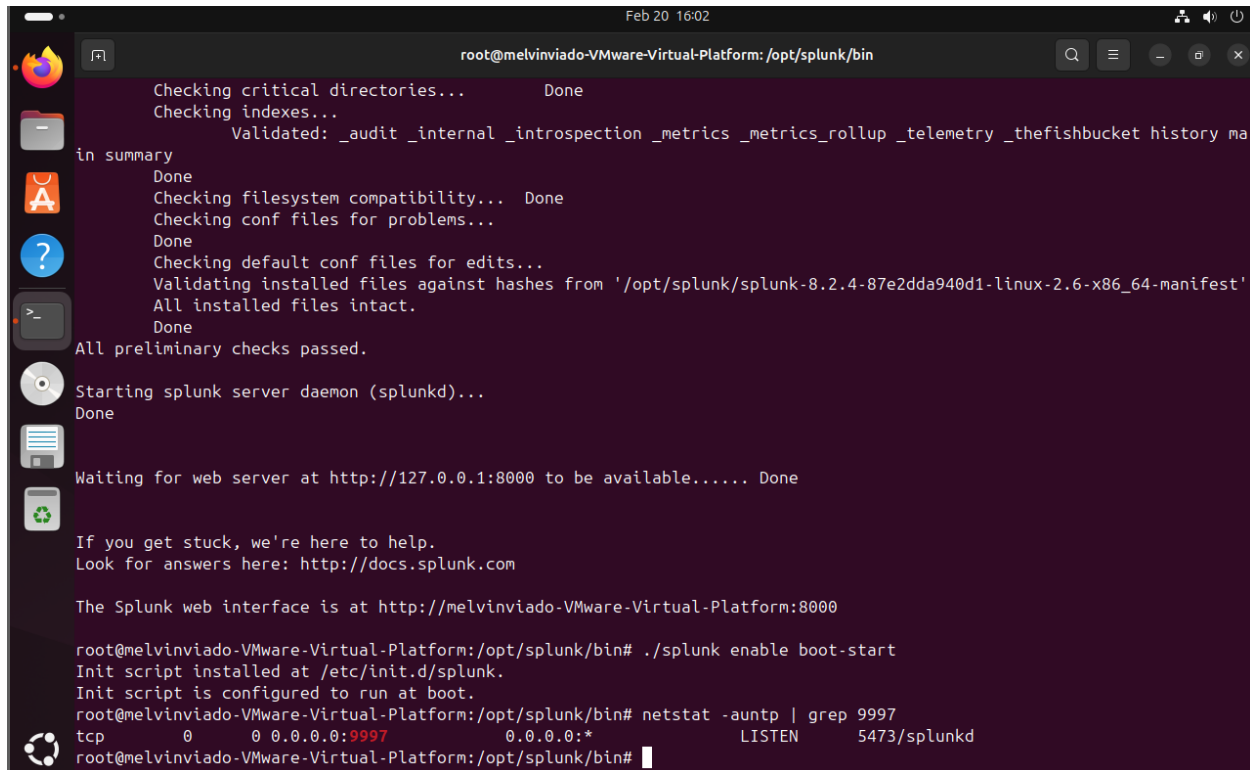
- a) Go to web interface of Splunk enterprise by using ipaddress:8000. It will ask your administrator username and password for Splunk. Go to settings tab above and click on forwarding and receiving
- b) Click on Configure receiving under the Receive data column.
- c) Click on new receiving ports
- d) Input the default receiving port on TCP port 9997

[Attach your Screenshot]



- e) Go back to your command line and run the **command** `netstat -aunt | grep 9997`. This would allow us to see the new configured receiving port we added in previous step.

[Attach your Screenshot]



```
Feb 20 16:02
root@melvinviado-VMware-Virtual-Platform: /opt/splunk/bin

Checking critical directories... Done
Checking indexes...
Validated: _audit _internal _introspection _metrics _metrics_rollup _telemetry _thefishbucket history ma
in summary
Done
Checking filesystem compatibility... Done
Checking conf files for problems...
Done
Checking default conf files for edits...
Validating installed files against hashes from '/opt/splunk/splunk-8.2.4-87e2dda940d1-linux-2.6-x86_64-manifest'
All installed files intact.
Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
Done

Waiting for web server at http://127.0.0.1:8000 to be available..... Done

If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://melvinviado-VMware-Virtual-Platform:8000

root@melvinviado-VMware-Virtual-Platform:/opt/splunk/bin# ./splunk enable boot-start
Init script installed at /etc/init.d/splunk.
Init script is configured to run at boot.
root@melvinviado-VMware-Virtual-Platform:/opt/splunk/bin# netstat -aunt | grep 9997
tcp        0      0 0.0.0.0:9997          0.0.0.0:*             LISTEN     5473/splunkd
root@melvinviado-VMware-Virtual-Platform:/opt/splunk/bin#
```

Now our Splunk is listening on the port 9997 for any logs from our universal forwarder. **Congratulations, you have just configured Splunk Ubuntu Server to receive logs now.**

That's the end of installing Splunk. Next, would be to install our Splunk universal forwarder to forward logs to Splunk.

Step 3: Download and install Splunk Universal Forwarder on second ubuntu machine

- a)** Go to your second ubuntu machine you installed, open the terminal window and change to the downloads directory using the command `cd Downloads` to run the command (I believe you all should know by now what `cd` (change directory) command means)

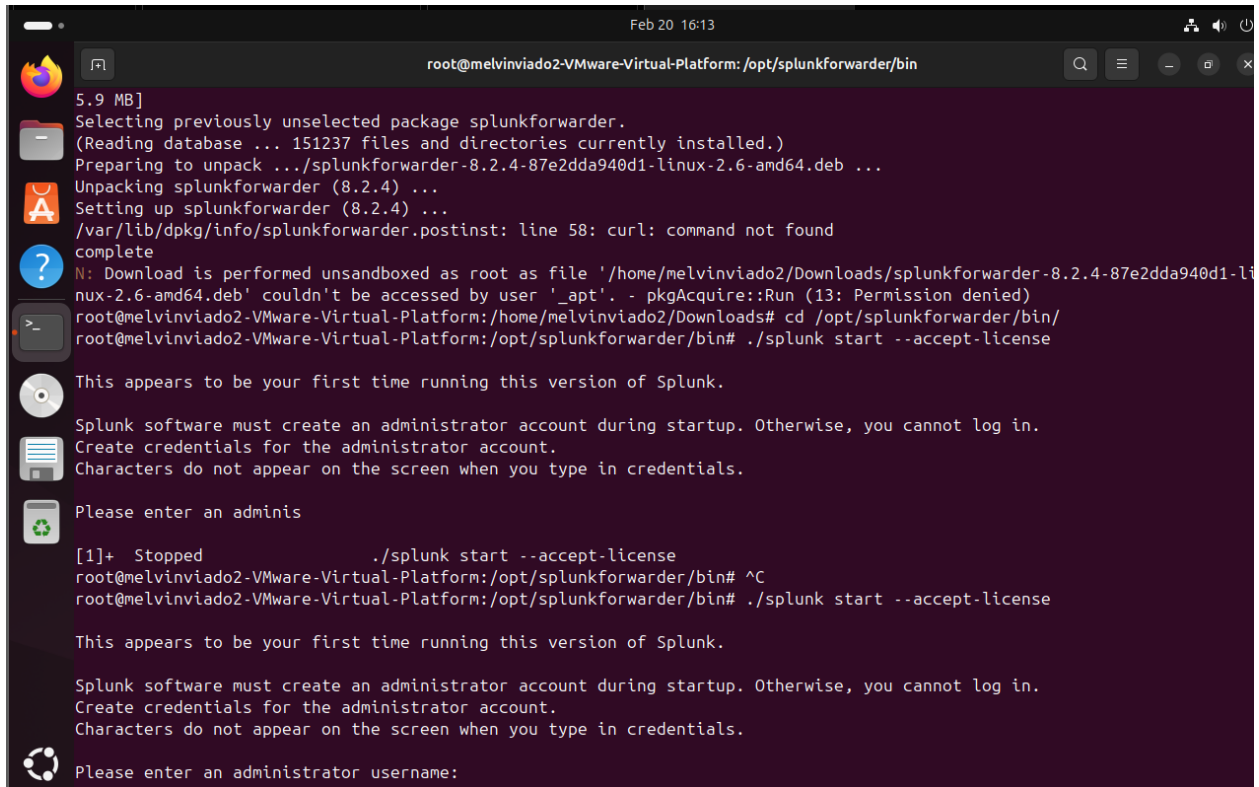
```
wget -O splunkforwarder-8.2.4-87e2dda940d1-linux-2.6-amd64.deb
```

```
'https://download.splunk.com/products/universalforwarder/releases/8.2.4/linux/splunkforwarder-8.2.4-87e2dda940d1-linux-2.6-amd64.deb'
```

- b)** Run the command `sudo su` and input password to change to root user.
- c)** Install the universal forwarder using the command.

```
apt install ./splunkforwarder-8.2.4-87e2dda940d1-linux-2.6-amd64.deb
```

- d) Run the command `cd /opt/splunkforwarder/bin/` to change to the Splunk forwarder bin directory and start the license using the command `./splunk start --accept-license` to accept the license agreement **[Attach your Screenshot]**



The screenshot shows a terminal window with the following content:

```
Feb 20 16:13
root@melvinviado2-VMware-Virtual-Platform: /opt/splunkforwarder/bin

5.9 MB]
Selecting previously unselected package splunkforwarder.
(Reading database ... 151237 files and directories currently installed.)
Preparing to unpack .../splunkforwarder-8.2.4-87e2dda940d1-linux-2.6-amd64.deb ...
Unpacking splunkforwarder (8.2.4) ...
Setting up splunkforwarder (8.2.4) ...
/var/lib/dpkg/info/splunkforwarder.postinst: line 58: curl: command not found
complete
N: Download is performed unsandboxed as root as file '/home/melvinviado2/Downloads/splunkforwarder-8.2.4-87e2dda940d1-linux-2.6-amd64.deb' couldn't be accessed by user '_apt'. - pkgAcquire::Run (13: Permission denied)
root@melvinviado2-VMware-Virtual-Platform:/home/melvinviado2/Downloads# cd /opt/splunkforwarder/bin/
root@melvinviado2-VMware-Virtual-Platform:/opt/splunkforwarder/bin# ./splunk start --accept-license

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an adminis

[1]+  Stopped                  ./splunk start --accept-license
root@melvinviado2-VMware-Virtual-Platform:/opt/splunkforwarder/bin# ^C
root@melvinviado2-VMware-Virtual-Platform:/opt/splunkforwarder/bin# ./splunk start --accept-license

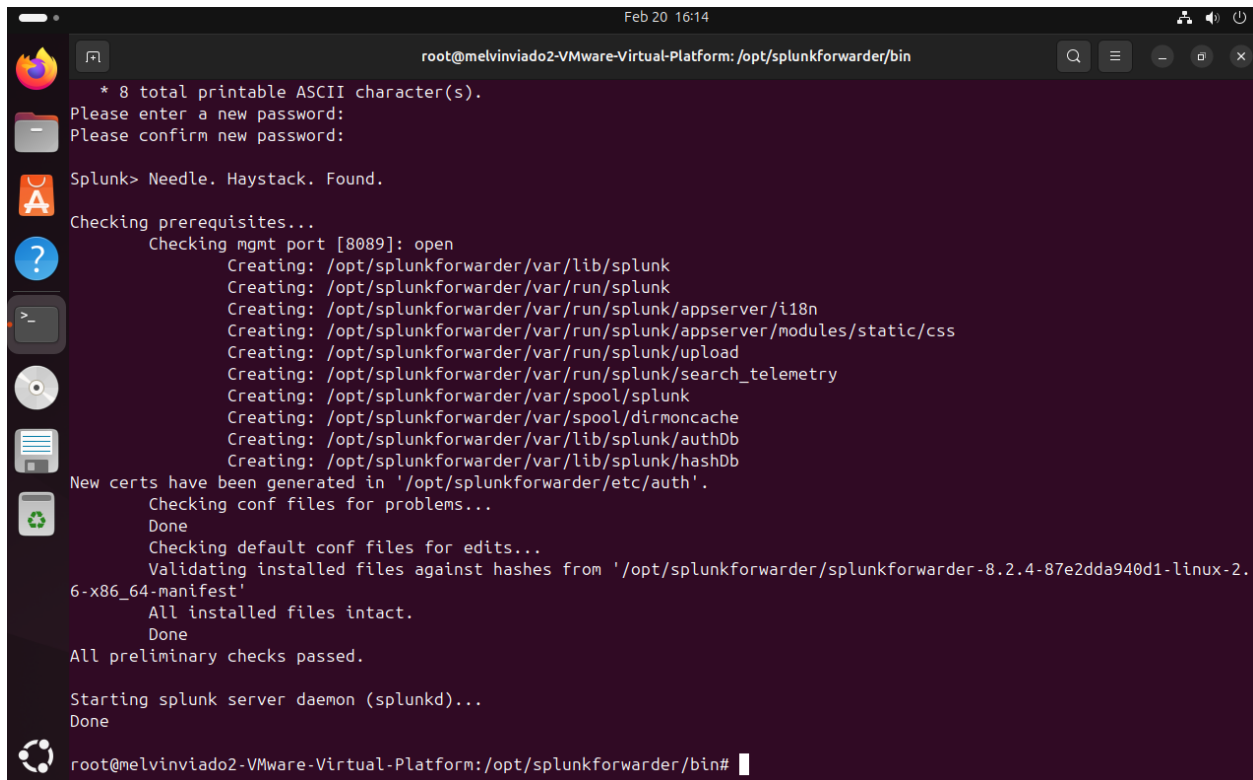
This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username:
```

e) set your forwarders username to `firstname-NETW2200` and input password

[Attach your Screenshot]



```
Feb 20 16:14
root@melvinviado2-VMware-Virtual-Platform: /opt/splunkforwarder/bin

* 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:

Splunk> Needle. Haystack. Found.

Checking prerequisites...
  Checking mgmt port [8089]: open
    Creating: /opt/splunkforwarder/var/lib/splunk
    Creating: /opt/splunkforwarder/var/run/splunk
    Creating: /opt/splunkforwarder/var/run/splunk/appserver/i18n
    Creating: /opt/splunkforwarder/var/run/splunk/appserver/modules/static/css
    Creating: /opt/splunkforwarder/var/run/splunk/upload
    Creating: /opt/splunkforwarder/var/run/splunk/search_telemetry
    Creating: /opt/splunkforwarder/var/spool/splunk
    Creating: /opt/splunkforwarder/var/spool/dirmoncache
    Creating: /opt/splunkforwarder/var/lib/splunk/authDb
    Creating: /opt/splunkforwarder/var/lib/splunk/hashDb
  New certs have been generated in '/opt/splunkforwarder/etc/auth'.
    Checking conf files for problems...
    Done
    Checking default conf files for edits...
    Validating installed files against hashes from '/opt/splunkforwarder/splunkforwarder-8.2.4-87e2dda940d1-linux-2.6-x86_64-manifest'
    All installed files intact.
    Done
  All preliminary checks passed.

  Starting splunk server daemon (splunkd)...
  Done

root@melvinviado2-VMware-Virtual-Platform: /opt/splunkforwarder/bin#
```

- f) Next step would be to add our Splunk's IP address to our forwarder to forward logs using the command `./splunk add forward-server (your splunk IP-address:port added to splunk to receive logs)`

e.g. `./splunk add forward-server 192.168.80.130:9997`

[Attach your Screenshot]

```

Feb 20 16:38
root@melvinviado2-VMware-Virtual-Platform: /opt/splunkforwarder/bin

Cannot initialize: /opt/splunkforwarder/etc/system/metadata/local.meta: Permission denied
Cannot initialize: /opt/splunkforwarder/etc/apps/learned/metadata/local.meta: Permission denied
Error opening username mapping file: /opt/splunkforwarder/etc/users/users.ini err: Cannot open file=/opt/splunkforwarder/etc/users/users.ini for parsing: Permission denied
Cannot initialize: /opt/splunkforwarder/etc/system/metadata/local.meta: Permission denied
Cannot initialize: /opt/splunkforwarder/etc/apps/learned/metadata/local.meta: Permission denied
Error opening username mapping file: /opt/splunkforwarder/etc/users/users.ini err: Cannot open file=/opt/splunkforwarder/etc/users/users.ini for parsing: Permission denied
Cannot initialize: /opt/splunkforwarder/etc/system/metadata/local.meta: Permission denied
Not found required stanza [general] in /opt/splunkforwarder/etc/instance.cfg; latter corrupt
Failed to open splunk.secret '/opt/splunkforwarder/etc/auth/splunk.secret' file. Some passwords will not work. error=Permission denied
Cannot load to modify: /opt/splunkforwarder/etc/system/metadata/local.meta
/opt/splunkforwarder/etc/system/local: Setting /nobody/system/server/sslConfig/sslPassword = $7$c/C/zRgLNx5Jyc7ehmZFwiiQs3tLPrQujtK175GevwzMGAr+P29UDA==: Cannot load ini file to modify
Cannot load to modify: /opt/splunkforwarder/etc/system/metadata/local.meta
/opt/splunkforwarder/etc/system/local: Setting /nobody/system/server/sslConfig/sslPassword = $7$eozb/ZdP3h1IMSikUV1XuxInGC0jbs4Dk0AynTZAB8fgEULL9DD0oQ==: Cannot load ini file to modify
Pid file "/opt/splunkforwarder/var/run/splunk/splunkd.pid" unreadable.: Permission denied
Operation "mkdir(2)" failed in /opt/splunk/src/libzero/conf-mutator-locking.c:93, ensure_exists_directory(); Permission denied
melvinviado2@melvinviado2-VMware-Virtual-Platform: /opt/splunkforwarder/bin$ sudo su
[sudo] password for melvinviado2:
root@melvinviado2-VMware-Virtual-Platform: /opt/splunkforwarder/bin# ./splunk add forward-server 192.168.147.129:9997
splunkd 3853 was not running.
Stopping splunk helpers...

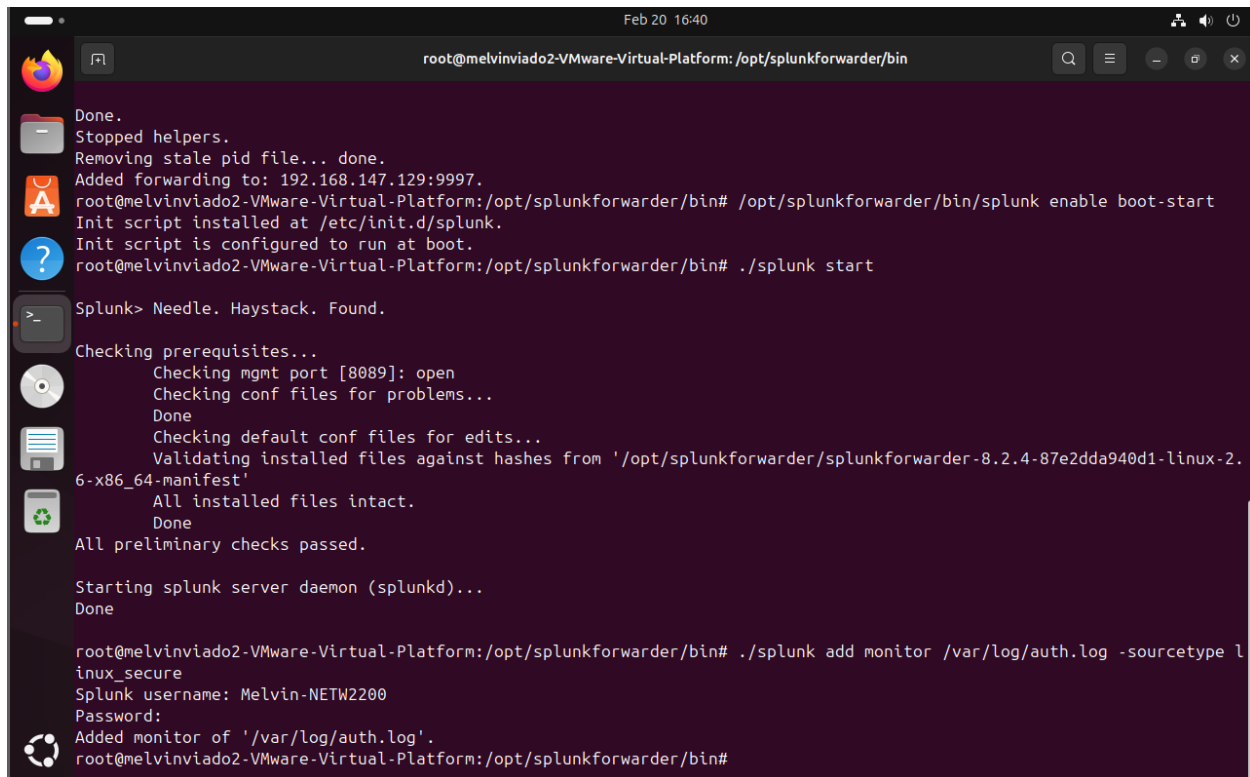
Done.
Stopped helpers.
Removing stale pid file... done.
Added forwarding to: 192.168.147.129:9997.
root@melvinviado2-VMware-Virtual-Platform: /opt/splunkforwarder/bin#

```

- g) Run the command to enable boot at start
`/opt/splunkforwarder/bin/splunk enable boot-start`
- h) Next would be to run the command `./splunk start` on the universal forwarder to make sure we start the service.
- i) Next step would be to run the command `./splunk add monitor /var/log/auth.log -sourcetype linux_secure` to send our

authentication logs to Splunk (**Note:** This may ask for your Splunk username and password to authenticate)

[Attach your Screenshot]



```
root@melvinviado2-VMware-Virtual-Platform: /opt/splunkforwarder/bin
Done.
Stopped helpers.
Removing stale pid file... done.
Added forwarding to: 192.168.147.129:9997.
root@melvinviado2-VMware-Virtual-Platform:/opt/splunkforwarder/bin# /opt/splunkforwarder/bin/splunk enable boot-start
Init script installed at /etc/init.d/splunk.
Init script is configured to run at boot.
root@melvinviado2-VMware-Virtual-Platform:/opt/splunkforwarder/bin# ./splunk start

Splunk> Needle. Haystack. Found.

Checking prerequisites...
Checking mgmt port [8089]: open
Checking conf files for problems...
Done
Checking default conf files for edits...
Validating installed files against hashes from '/opt/splunkforwarder/splunkforwarder-8.2.4-87e2dda940d1-linux-2.6-x86_64-manifest'
All installed files intact.
Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
Done

root@melvinviado2-VMware-Virtual-Platform:/opt/splunkforwarder/bin# ./splunk add monitor /var/log/auth.log -sourcetype linux_secure
Splunk username: Melvin-NETW2200
Password:
Added monitor of '/var/log/auth.log'.
root@melvinviado2-VMware-Virtual-Platform:/opt/splunkforwarder/bin#
```

Step 4: Searching and Reporting App – Capture the logs in Splunk Ubuntu Server

- a) Once we run add-monitor command in previous, the next thing we would do is to wait for few minutes for this to take effect. After few minutes, click on Data summary on the search page of Ubuntu Splunk Server and you would see your authentication logs data has started ingesting into Splunk as seen below. Keep time picker as 24 hours.

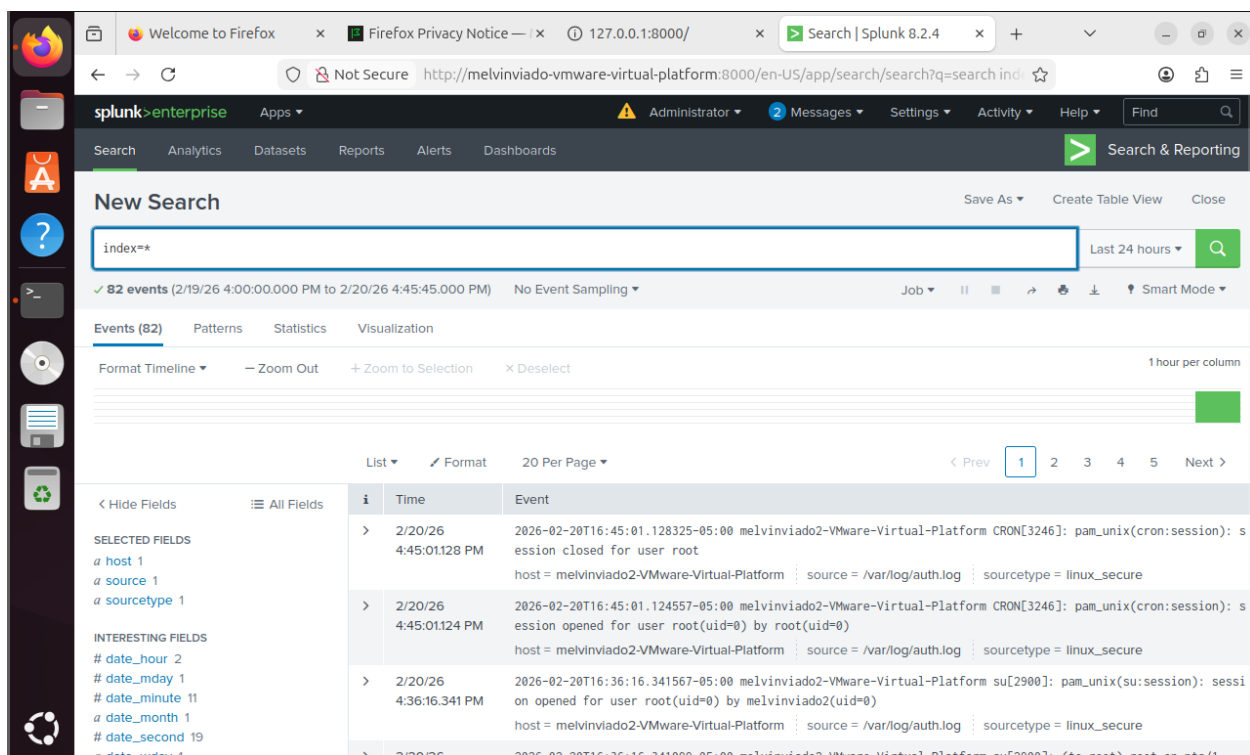
[Attach your Screenshot]

The screenshot displays the Splunk Search & Reporting interface. The search bar contains the query `source="/var/log/auth.log"` and the time range is set to "Last 24 hours". The search results show 82 events. The table below lists the events:

Time	Event
2/20/26 4:45:01.128 PM	2026-02-20T16:45:01.128325-05:00 melvinviado2-VMware-Virtual-Platform CRON[3246]: pam_unix(cron:session): session closed for user root host = melvinviado2-VMware-Virtual-Platform source = /var/log/auth.log sourcetype = linux_secure
2/20/26 4:45:01.124 PM	2026-02-20T16:45:01.124557-05:00 melvinviado2-VMware-Virtual-Platform CRON[3246]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0) host = melvinviado2-VMware-Virtual-Platform source = /var/log/auth.log sourcetype = linux_secure
2/20/26 4:36:16.341 PM	2026-02-20T16:36:16.341567-05:00 melvinviado2-VMware-Virtual-Platform su[2900]: pam_unix(su:session): session opened for user root(uid=0) by melvinviado2(uid=0) host = melvinviado2-VMware-Virtual-Platform source = /var/log/auth.log sourcetype = linux_secure
2/20/26	2026-02-20T16:36:16.341099-05:00 melvinviado2-VMware-Virtual-Platform su[2900]: (to root) root on pts/1

- b) Now we run the command `index=*`, this way we would be able to see some data in our Splunk (To make the most of our free Splunk license in this lab, we'll send only our authentication logs to Splunk for the time being, and we'll explore with more log forwarding in future Labs if we need more detections.)

[Attach your Screenshot]



The screenshot shows the Splunk Search interface. The search bar contains the query `index=*`. The results show 82 events. The first three events are listed below:

Time	Event
2/20/26 4:45:01.128 PM	2026-02-20T16:45:01.128325-05:00 melvinviado2-VMware-Virtual-Platform CRON[3246]: pam_unix(cron:session): session closed for user root host = melvinviado2-VMware-Virtual-Platform source = /var/log/auth.log sourcetype = linux_secure
2/20/26 4:45:01.124 PM	2026-02-20T16:45:01.124557-05:00 melvinviado2-VMware-Virtual-Platform CRON[3246]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0) host = melvinviado2-VMware-Virtual-Platform source = /var/log/auth.log sourcetype = linux_secure
2/20/26 4:36:16.341 PM	2026-02-20T16:36:16.341567-05:00 melvinviado2-VMware-Virtual-Platform su[2900]: pam_unix(su:session): session opened for user root(uid=0) by melvinviado2(uid=0) host = melvinviado2-VMware-Virtual-Platform source = /var/log/auth.log sourcetype = linux_secure

Congratulations, you've just set up Splunk and the Splunk universal forwarder to forward your authentication logs from Ubuntu Linux to the SIEM (Splunk)

Questions

Answer the following questions using your search you ran above from the command `index=* search`.

- What is your Splunk index? (3marks)

/var/log/auth.log

- What is your splunk_server name? (1mark)

Melvinviado2-VMware-Virtual-Platform

- What does the * in the `index=*` means? (Hint: it's one word) (1 mark)

all

- What type of logs are we currently collecting in Splunk? (1 mark)

Authorization logs

- Based on your screenshot for the search (`index=*`), how many events can you see? (1mark)

82 events

Things to Explore:

You are welcome to explore beyond the mandatory requirements if you wish.

General submission requirements

- Include an opening comment with your full name, date, and a short description.
- **Marks distribution:**
 - a) 16 Screenshots each 1 mark = 16 marks
 - b) Questions = 7 marks
 - c) Video Demonstration = 7 marks
- **Do not alter the sequence of steps of this document. Do not delete the words [Attach the screenshot] anywhere from the assignment. Keep the numbering same. Paste the screenshots wherever it's being asked- do not use screenshots of professors from lab manual. They are for your reference only and would be considered an Academic Integrity Breach. You should be not even submitting another student's screenshot- Academic Integrity alert.**
- **Submit a video recording (sharing screen) briefly explaining the steps taken to perform the lab. If the video recording stating the above requirements are not submitted, you will be eligible to get marks out of 60% of Lab2 marks.**

*****GOOD LUCK*****